# Experiment 09:

## 1. Displaying Hping3 Help Menu

```
student@Shivam:~$ hping3 -h
usage: hping3 host [options]
  -h  --help      show this help
  -v  --version   show version
  -c  --count     packet count
  -i  --interval  wait (uX for X microseconds, for example -i u1000)
      --fast      alias for -i u10000 (10 packets for second)
      --faster    alias for -i u1000 (100 packets for second)
      --flood      sent packets as fast as possible. Don't show replies.
  -n  --numeric   numeric output
  -q  --quiet     quiet
  -I  --interface interface name (otherwise default routing interface)
  -V  --verbose   verbose mode
  -D  --debug     debugging info
  -z  --bind      bind ctrl+z to ttl           (default to dst port)
  -Z  --unbind    unbind ctrl+z
      --beep      beep for every matching packet received
Mode
  default mode     TCP
  -0  --rawip      RAW IP mode
  -1  --icmp       ICMP mode
  -2  --udp        UDP mode
  -8  --scan       SCAN mode.
                   Example: hping --scan 1-30,70-90 -S www.target.host
  -9  --listen     listen mode
IP
  -a  --spoof      spoof source address
  --rand-dest      random destionation address mode. see the man.
  --rand-source    random source address mode. see the man.
  -t  --ttl        ttl (default 64)
  -N  --id         id (default random)
  -W  --winid      use win* id byte ordering
  -r  --rel        relativize id field          (to estimate host traffic)
  -f  --frag       split packets in more frag.  (may pass weak acl)
  -x  --morefrag   set more fragments flag
  -y  --dontfrag   set don't fragment flag
```

**Observation:**

The hping3 -h command was executed to display the help menu, which lists the available options, modes and parameters used for network packet testing.

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## 2. Launching Wireshark

```
student@Shivam:~$ sudo wireshark
[sudo] password for student:
** (wireshark:5770) 11:35:31.091370 [GUI WARNING] -- QStandardPaths: XDG_RUNTIME_DI
```

**Observation:**

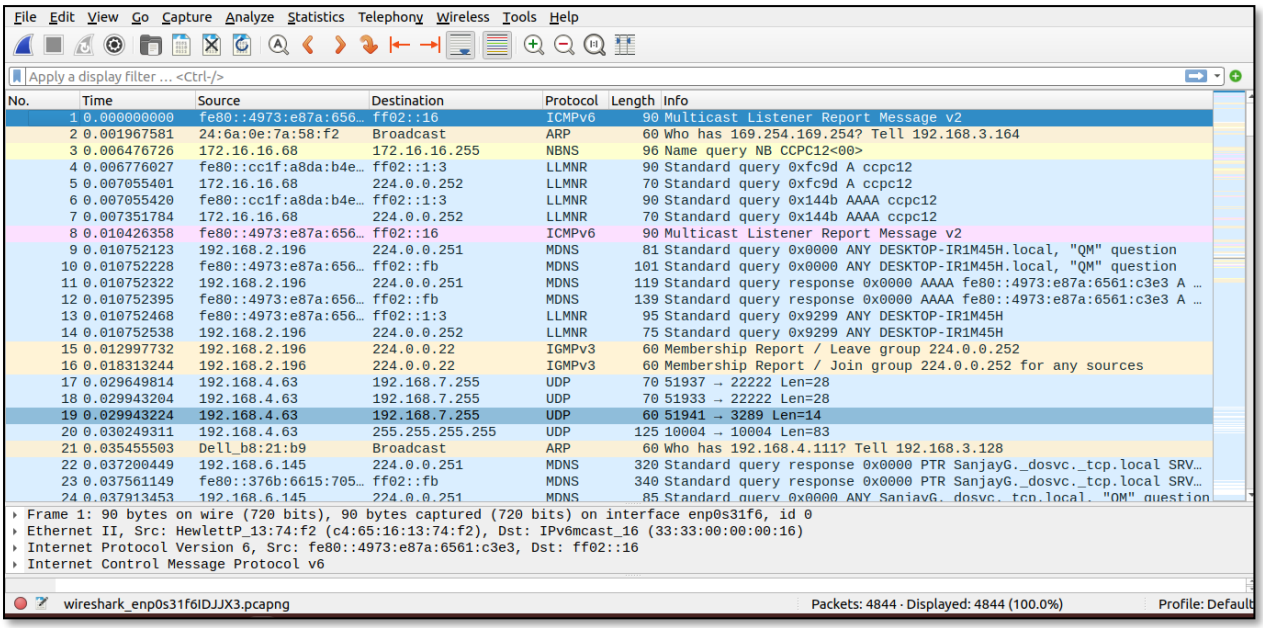
The sudo wireshark command was executed to launch Wireshark with administrative privileges for capturing and analyzing network packets.

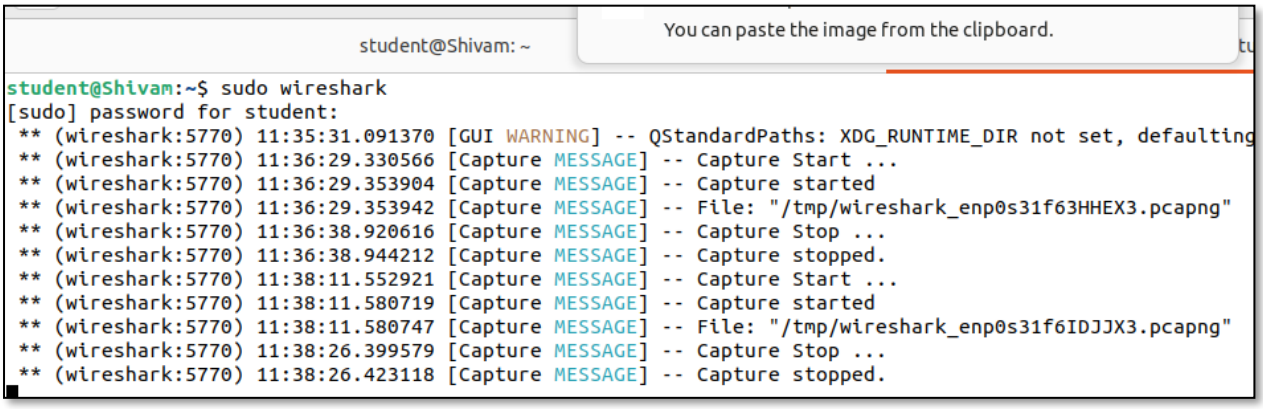
## 3. DNS Resolution and Initial Packet Capture

```
student@Shivam:~$ nslookup vulnweb.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   vulnweb.com
Address: 44.228.249.3
Name:   vulnweb.com
Address: 64:ff9b::2ce4:f903
```

```
student@Shivam:~$ sudo hping3 44.228.249.3
HPING 44.228.249.3 (enp0s31f6 44.228.249.3): NO FLAGS are set, 40 headers + 0 data bytes
```




## Observation:

- The nslookup vulweb.com command was used to resolve the domain name and obtain its IPv4 and IPv6 addresses.
- The hping3 44.228.249.3 command was executed to send TCP packets to the resolved IPv4 address.
- Wireshark was used to capture and display network traffic, and the terminal shows the capture process being started and stopped.

- The Hping3 statistics show 210 packets transmitted, 0 packets received, and 100% packet loss.

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## 4. Filtering Network Traffic by IP Address

```
student@Shivam:~$ sudo hping3 44.228.249.3
HPING 44.228.249.3 (enp0s31f6 44.228.249.3): NO FLAGS are set, 40 headers + 0 data bytes
^C
--- 44.228.249.3 hping statistic ---
210 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

ip.addr==44.228.249.3

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 170 | 0.604499717 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1833 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 645 | 1.604565622 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1834 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 876 | 2.604624435 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1835 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 1130 | 3.604688683 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1836 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 1341 | 4.604746708 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1837 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 1516 | 5.604803373 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1838 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 1853 | 6.604855747 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1839 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 2223 | 7.604901017 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1840 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 2580 | 8.604958569 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1841 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 2865 | 9.605013960 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1842 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 3489 | 10.605077167 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1843 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 3814 | 11.605148815 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1844 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 4202 | 12.605223583 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1845 → 0 [<None>] Seq=1 Win=512 Len=0 |
| 4483 | 13.605318264 | 192.168.6.164 | 44.228.249.3 | TCP | 54 | 1846 → 0 [<None>] Seq=1 Win=512 Len=0 |

### Observation:

- The Hping3 command was executed against IP address 44.228.249.3, and the output shows 210 packets transmitted, 0 packets received, and 100% packet loss.
- The Wireshark display filter ip.addr==44.228.249.3 was applied to display packets associated with the specified IP address, showing TCP traffic from the local system to the target.

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## 5. Sending ICMP Packets and Testing Transmission Speed

ip.addr==44.228.249.3

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 7757 | 26.893391803 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=9728/38, ttl=64 (no response fou… |
| 7772 | 26.993457344 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=9984/39, ttl=64 (no response fou… |
| 7809 | 27.093508979 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=10240/40, ttl=64 (no response fo… |
| 7844 | 27.193561364 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=10496/41, ttl=64 (no response fo… |
| 7865 | 27.293608115 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=10752/42, ttl=64 (no response fo… |
| 7890 | 27.393657108 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=11008/43, ttl=64 (no response fo… |
| 7924 | 27.493719277 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=11264/44, ttl=64 (no response fo… |
| 7948 | 27.593769325 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=11520/45, ttl=64 (no response fo… |
| 7988 | 27.693821862 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=11776/46, ttl=64 (no response fo… |
| 8017 | 27.793881410 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=12032/47, ttl=64 (no response fo… |
| 8030 | 27.893950117 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=12288/48, ttl=64 (no response fo… |
| 8050 | 27.994011733 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=12544/49, ttl=64 (no response fo… |
| 8074 | 28.094051958 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=12800/50, ttl=64 (no response fo… |
| 8098 | 28.194081062 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=13056/51, ttl=64 (no response fo… |
| 8114 | 28.294112348 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=13312/52, ttl=64 (no response fo… |
| 8135 | 28.394169325 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=13568/53, ttl=64 (no response fo… |
| 8172 | 28.494226692 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=13824/54, ttl=64 (no response fo… |
| 8187 | 28.594270487 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=14080/55, ttl=64 (no response fo… |
| 8206 | 28.694297728 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=14336/56, ttl=64 (no response fo… |
| 8234 | 28.794323585 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=14592/57, ttl=64 (no response fo… |
| 8248 | 28.894363426 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=14848/58, ttl=64 (no response fo… |
| 8262 | 28.994412924 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=15104/59, ttl=64 (no response fo… |
| 8292 | 29.094462968 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0x221c, seq=15360/60, ttl=64 (no response fo… |

```
▸ Frame 6649: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface enp0s31f6, id 0
▸ Ethernet II, Src: c8:7f:54:16:b9:13 (c8:7f:54:16:b9:13), Dst: Sophos_c8:66:82 (7c:5a:1c:c8:66:82)
▸ Internet Protocol Version 4, Src: 192.168.6.164, Dst: 44.228.249.3
```

```
student@Shivam:~$ sudo hping3 44.228.249.3 -1
HPING 44.228.249.3 (enp0s31f6 44.228.249.3): icmp mode set, 28 headers + 0 data bytes
^C
--- 44.228.249.3 hping statistic ---
49 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

**Observation:**

- The hping3 44.228.249.3 -1 command was executed in ICMP mode, and the output shows 49 packets transmitted, 0 packets received, and 100% packet loss.
- Wireshark captured and displayed ICMP Echo Request packets sent to the target IP address without corresponding replies.

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## 6. Monitoring ICMP Traffic in Wireshark

```
student@Shivam:~$ sudo hping3 44.228.249.3 -1 --fast
HPING 44.228.249.3 (enp0s31f6 44.228.249.3): icmp mode set, 28 headers + 0 data bytes
^C
--- 44.228.249.3 hping statistic ---
387 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

ip.addr==44.228.249.3

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 1193 | 4.916698343 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1213 | 5.016849802 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1228 | 5.116996279 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1265 | 5.217063383 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1315 | 5.317131673 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1329 | 5.417203493 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1361 | 5.517347562 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1397 | 5.617447258 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1444 | 5.717496084 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1474 | 5.817570785 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1509 | 5.917643100 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1529 | 6.017786231 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1546 | 6.117862118 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1588 | 6.217928908 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1611 | 6.317997824 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1635 | 6.418058027 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1666 | 6.518105793 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1693 | 6.618148677 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1708 | 6.718219992 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1736 | 6.818274266 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1762 | 6.918318582 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1802 | 7.018392968 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1823 | 7.118461087 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |

**Observation:**

- The hping3 44.228.249.3 -1 --fast command was executed to send ICMP packets at a faster rate. The output shows 387 packets transmitted, 0 packets received, and 100% packet loss.
- Wireshark displayed the captured ICMP Echo Request packets directed to the target IP address, with no corresponding replies visible in the capture.

## 7. Using the Faster Packet Transmission Option

```
student@Shivam:~$ sudo hping3 44.228.249.3 -1 --faster
HPING 44.228.249.3 (enp0s31f6 44.228.249.3): icmp mode set, 28 headers + 0 data bytes
^C
--- 44.228.249.3 hping statistic ---
655631 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

ip.addr==44.228.249.3

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 1815… | 6.371535682 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.371537642 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.371539691 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.371541630 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.371543809 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.371546052 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.371548691 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.372896584 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.372898879 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.372901271 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.372903644 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.372905967 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.372908341 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.372910456 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.372913094 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.373144652 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.373147172 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.373149459 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.373151882 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.373154129 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.373156677 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.373159158 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |
| 1815… | 6.373161375 | 192.168.6.164 | 44.228.249.3 | ICMP | 42 | Echo (ping) request id=0 |

```
▸ Frame 2: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface enp0s31f6, id 0
▸ Ethernet II, Src: c8:7f:54:16:b9:13 (c8:7f:54:16:b9:13), Dst: Sophos_c8:66:82 (7c:5a:1c:c8:66:82)
▸ Internet Protocol Version 4, Src: 192.168.6.164, Dst: 44.228.249.3
```

### Observation:

- The hping3 44.228.249.3 -1 --faster command was executed to increase the ICMP packet transmission rate. The output shows 655631 packets transmitted, 0 packets received, and 100% packet loss.
- Wireshark captured and displayed the corresponding ICMP Echo Request packets sent from the local system to the target IP address.

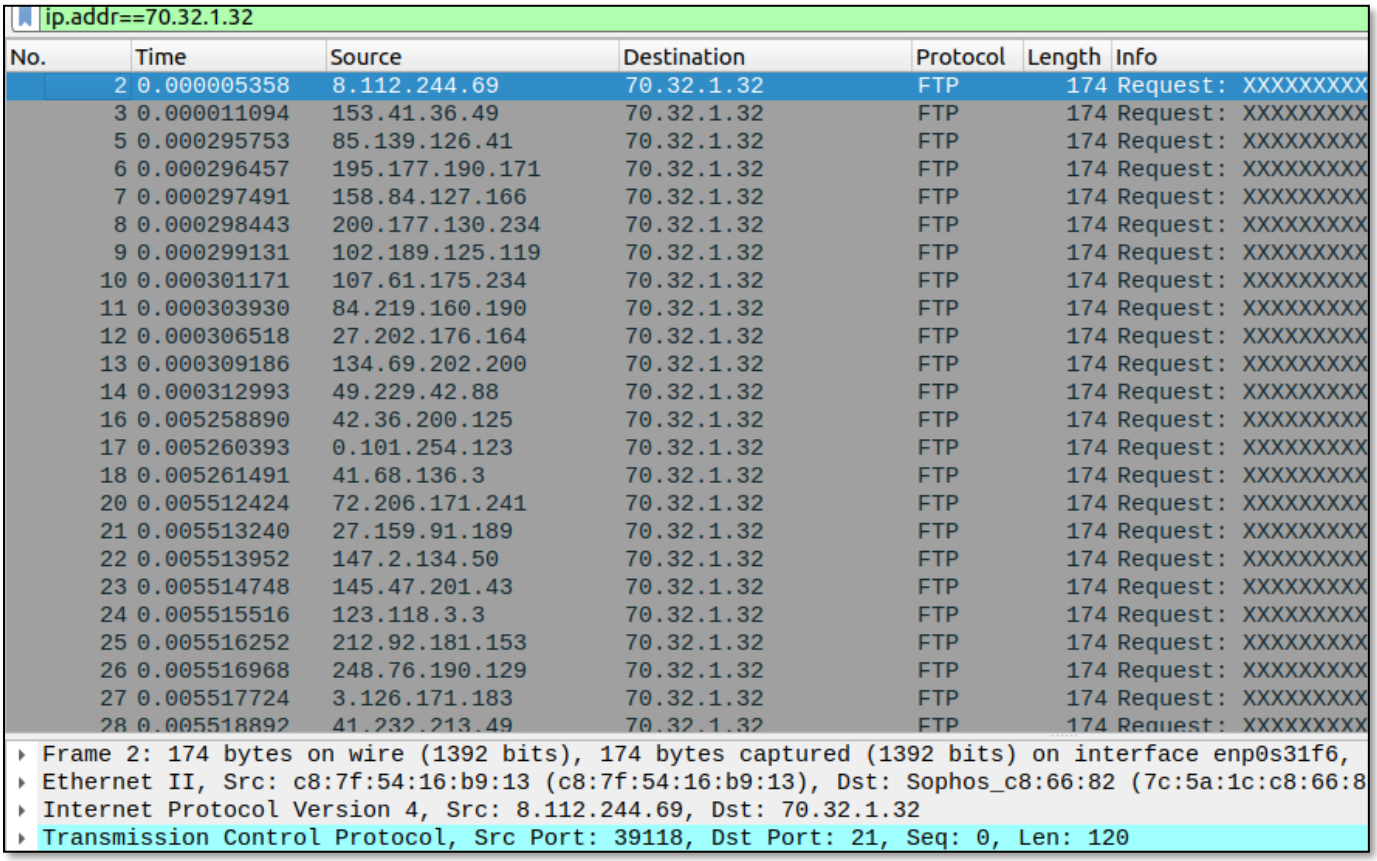
---

## 8. DNS Lookup Failure, Flood Mode, and TCP Traffic Analysis

```
student@Shivam:~$ nslookup www.hping3testsite.com
Server:         127.0.0.53
Address:        127.0.0.53#53

** server can't find www.hping3testsite.com: NXDOMAIN
```

```
student@Shivam:~$ sudo hping3 -c 10000 -d 120 -S -w 64 -p 21 --flood --rand-source 70.32.1.32
HPING 70.32.1.32 (enp0s31f6 70.32.1.32): S set, 40 headers + 120 data bytes
hping in flood mode, no replies will be shown
^C
--- 70.32.1.32 hping statistic ---
458717 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```




**Observation:**

- The nslookup www.hping3testsite.com command returned NXDOMAIN, indicating that the domain name could not be resolved. An Hping3 command using the --flood and --rand-source options was then executed against IP address 70.32.1.32. The output shows 458717 packets transmitted, 0 packets received, and 100% packet loss.
- Wireshark displayed TCP packets directed to IP address 70.32.1.32, with destination port 21, which is associated with FTP traffic. The capture shows repeated FTP requests, but it does not establish whether a DoS attack was successful.